Indian Institute Of Information Technology – Vadodara

Embedded System Project Summary

Group – 9:

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Embedded system for Nutrient Stress detection

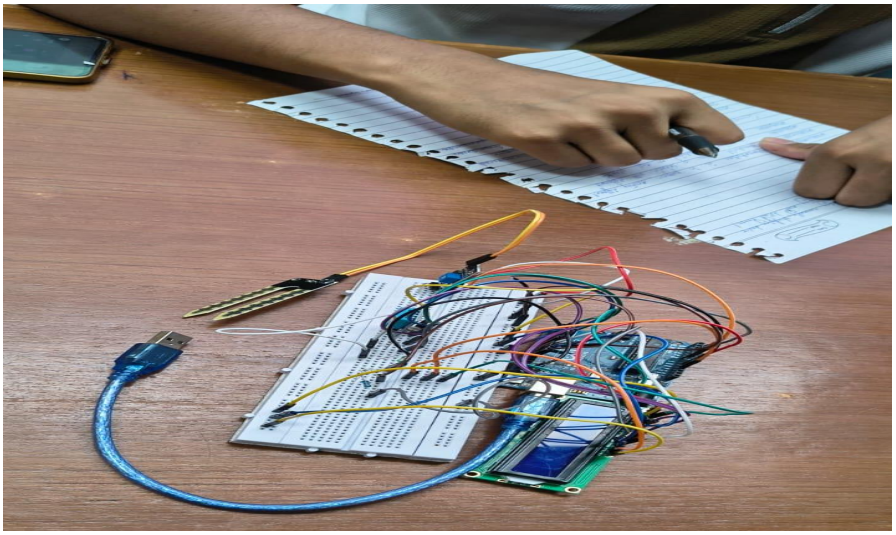
Work Completed Till Now:

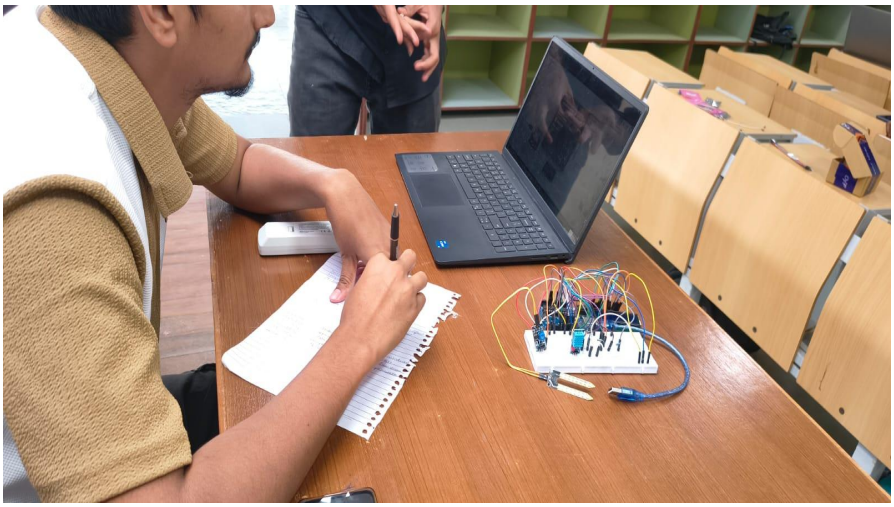
Hardware Implementation

In this phase, the hardware setup of the project was completed.
All required components such as Arduino Uno, soil moisture sensor, DHT11 sensor, LDR, LCD display, and LED were connected using a breadboard.

The circuit was first designed and verified in Tinkercad simulation. Based on the simulation results, the same connections were implemented in real hardware.

The system is designed to monitor environmental parameters like soil moisture, temperature, humidity, and light intensity to detect nutrient stress conditions.





Software Implementation (Embedded System)

The Arduino program for the system was developed.

The code reads data from all sensors and applies predefined threshold values to detect stress conditions. Based on the sensor values:

- Low moisture indicates water stress
- High temperature indicates heat stress
- Low light indicates light deficiency

The system then displays the plant condition on the LCD as:

- "Plant Healthy"
- "Nutrient Stress Detected"

An LED is used to indicate stress conditions.

The software logic has been prepared but is yet to be fully tested on real hardware.

Machine Learning / Deep Learning Module

To improve the accuracy of nutrient stress detection, a Machine Learning (Deep Learning) module was also implemented.

Work Done:

- Collected soil nutrient dataset (N, P, K, pH)
- Performed data preprocessing:
    - Removed unnecessary columns (e.g., District)
    - Normalized data using StandardScaler
- Created classification labels:

    - Normal
    - Deficient
    - Stress

CNN-LSTM Model Implementation

A hybrid **CNN-LSTM model** was developed for nutrient stress prediction.

Input Features:

- Nitrogen (N)
- Phosphorous (P)
- Potassium (K)
- pH

Model Architecture:

- CNN is used to extract feature relationships
- LSTM is used to capture patterns in data
- Combined model gives final classification

Output:

- Soil condition classified as:
    - Normal
    - Deficient
    - Stress

Model Prediction Example:

Input:
N = 80, P = 40, K = 45, pH = 6.5

Output:
Predicted Soil Condition: **Deficient**

Relation with Research Work:

The project is inspired by research work that uses IoT sensors along with CNN-LSTM models for nutrient prediction.

- Research approach: IoT + CNN-LSTM + optimization techniques
- Our approach:
    - Embedded system using Arduino

- Deep learning model developed separately

Difference:

The research work uses advanced sensors and real-time cloud integration, while our project focuses on a simplified and cost-effective system.

Current Status:

- Hardware setup completed
- Circuit connections verified
- Arduino code developed
- ML/DL model implemented and tested
- System ready for integration and testing

Work Yet to Be Done:

- Upload and test Arduino code on hardware
- Verify sensor outputs in real-time
- Integrate ML model conceptually with embedded system
- Debug and optimize system performance
- Collect final results and screenshots
- Complete final report and documentation

Conclusion:

The project has progressed significantly with both hardware and software development completed. The embedded system is ready for testing, and the Machine Learning model has been successfully implemented for nutrient stress prediction.

This combination of embedded systems and AI provides a strong foundation for developing smart agriculture solutions.